MILITARY SPECIFICATION

MINE, CHEMICAL AGENT, VX, M23;

INERT COMPONENTS

1. SCOPE

1.1 This specification covers the inert components of a 2 gallon chemical agent mine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-P-268	Paper, Kraft, Untreated, Wrapping.
UU-P-271	Paper, Wrapping, Waterproofed Kraft.
PPP-B-00566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-591	Boxes, Fiberboard, Wood-Cleated.
PPF-B-601	Boxes, Mood, Cleated-Flywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-665	Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676	Boxes, Set-Up, Paperboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-P-3420	- Packaging Materials, Volatile Corrosion Inhibitor, Treated, Opaque.

FSC 1345

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL37-1-58 - Mine, Chemical Agent, VX, M23, Inert Components.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL37-1-58 or on the subsidiary drawings shall conform to the specifications thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the mine shall conform to the specifications and drawings listed on Drawing DL37-1-58 and the subsidiary drawings.

3.2 Shutter. The shutter shall be in a position to fire the fuze when the arrow on the arming plug indicates "ARMED"; and shall prevent the fuze from firing when the arrow indicated “SAFE”. The shutter guide pin and shutter bearing pin subassemblies shall be riveted to withstand a load of at least 175 pounds when tested in accordance with 4.2.4.1.

3.3 Arming of the arming plug. The arming plug assembly (Dwg C37-1-27) shall arm when a torque of 5 ± 3 inch-pounds is applied to the setting knob when tested in accordance with 4.2.4.2, Breakaway torque of 20 inch-pounds maximum is permitted.

3.4 Mine handle assembly. The mine handle assembly (Dwg D37-1-34) shall suffer no permanent distortion nor show evidence of separating from the cover when a force of 75± 5 pounds is applied in accordance with 4.2.4.3.

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3.5 Pressure plate assembly load. The pressure plate assembly as shown on Drawing D37-1-58 shall require a dead load of over 60 pounds to cause the pressure plate and moisture seal to separate from the retaining ring and shall operate for a distance of 3/16 inch under a load of 150 ± 50 pounds when tested in accordance with 4.2,4.4.

3.6 Leakage.

3.6.1 Internal air pressure of the arming plug. The arming plug assemblies shall show no evidence of leakage when tested in accordance with 4.2.4.5.

3.6.2 Burster tube assembly. The burster tube assembly (Dwg C37-1-54) shall not leak helium at a rate greater than 1.0 x 10-6 atmospheric cubic centimeter per second when tested in accordance with 4.2.4.6.

3.6.3 Agent compartment. The agent compartment (Drawing D37-1-34) shall not leak helium at a rate greater than 1.0 x 10-6 atmospheric cubic centimeter per second when tested in accordance with 4.2.4.7. Vacuum in the agent compartment shall not exceed 15 inches of Hg at any time during processing of mine.

3.6.4 Void between pressure plate and fuze well. The void between the pressure plate and fuze well shall not leak when subjected to an air pressure of 3 ± 1/2 pounds per square inch (psi) for a period of 20 ± 5 seconds and tested in accordance with 4.2.4.8.

3.7 Burster cavity. The burster cavity shall conform to the maximum dimensions of the M38 burster when tested in accordance with 4.2.4.9.

3.8 Workmanship. The components and subassemblies shall be free from damage which might adversely affect the assembly or functioning of the mine and shall be free from foreign matter such as chips, dirt, rust and grease. The cleaning method used shall not injure or contaminate any of the parts by the cleaning agents used. The interior and exterior coatings specified on applicable drawings shall be applied in such manner as to insure an even distribution and a thorough coating. There shall be no metal working (permanent deformation) of walls of agent cavity after the helium leak test.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection,

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his

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own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection,

4.2.1 Lotting. A lot shall consist of the components and subassemblies of one interfix number, produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

4.2.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive tests. Sample components, subassemblies and the level B preparation for delivery shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.2.3.2 For critical defects. Each body, tube and plug in their respective lot shall be inspected for their critical characteristic listed in the classification of defects.

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4.2.3.3 Classification of defects.

(a) Body, mine assembly (Dwg D37-1-34).

Categories	Defects	Acceptance standards
Critical:		
1	Helium leakage	4.2.4.7
Major:	AQL 0.65 percent defective	
101	Adapter alignment incorrect (Dwg C37-1-41)	
102	Cover to body crimps not continuous	
	AQL 1.0 percent defective	
103	Adapter diameter (1.328) incorrect (Dwg C37-1-41)	
104	Adapter angles (36° and 30°) incorrect (Dwg C37-1-41)	
105	Dimension at intersection of angles incorrect (Dwg C37-1-41)	
106	Dimension 4.895 incorrect	
107	Body opening not completely covered by adapter	
108	Mine handle test failure	4.2.4.3
109	Protective finish missing or incorrect	
110	Flatness and perpendicularity of 2-1/16 - 1/16 inches flat incorrect (Dwg D37-1-35)	
111	Threads incorrect	
112	Dimension "X" incorrect	
113	Flatness of cover incorrect	
114	Workmanship (3.8)	

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4.2.3.3 Classification of defects (continued).

(b) Tube, burster, assembly (Dwg C37-1-54).

Categories	Defects	Acceptance standards
Critical:		
1	Helium leakage	4.2.4.6
Major:	AQL 0.65 percent defective	
101	Diameter 0.992 incorrect	
102	Inside diameter and straightness incorrect	
103	Eccentricity excessive (3 places)	
	AQL 1.0 percent defective	
104	Dimension 4.885 incorrect	
105	Dimension 0.296 incorrect	
106	Angles 45° and 53° incorrect	
107	External threads incorrect	
108	Internal threads incorrect	
109	Internal thread depth incorrect	
110	Protective finish missing or incorrect	
111	Snap ring groove location and dimension incorrect	
112	Undercut missing or incorrect	
113	Internal or external rust	
114	Workmanship (3.8)	
Minor:	AQL 4.0 percent defective	
201	Radii and chamfers missing or incorrect	
202	External thread location incorrect	

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4.2.3.3 Classification of defects (continued).

(c) Spring, retainer, fuze (Dwg C37-1-52).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter of wire incorrect
102	Free height incorrect
103	Diameters at extremities of spring incorrect
104	Hooked end incorrectly formed
105	Protective finish missing
106	Number of coils incorrect

(d) Plug, arming, assembly (Dwg C37-1-27).

Categories	Defects	Acceptance standards
Critical:		
1	Functioning of shutter	4.2.4.1
Major:	AQL 0.65 percent defective	
101	Shutter strength inadequate	4.2.4.1
102	Arming torque incorrect	4.2.4.2
103	Air leakage	4.2.4.5
104	Warning marking incorrect	
	AQL 1.0 percent defective	
105	Component missing	
106	Component incorrectly assembled	
107	Stenciling location incorrect	
108	Workmanship (3.8)	
Minor:	AQL 4.0 percent defective	
201	Lubricant missing	
202	Protective finish missing	

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4.2.3.3 Classification of defects (continued).

(e) Spring, coil (Drawing B 37-1-66).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Free length incorrect
102	Load at compressed length 1.372 incorrect
103	Load at compressed length 1.560 incorrect
104	Max solid height (coils closed) 0.960
105	Protective finish missing
106	Embrittlement not removed
107	Workmanship (3.8)

(f) Mine, chemical agent, VX, M23 - inert components (Dwg D37-1-58).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.65 percent defective	
101	Load test failure	4.2.4.4
102	Air test failure	4.2.4.8
103	Burster well adapter alignment incorrect	
104	Burster cavity incorrect	4.2.4.9
	AQL 1.0 percent defective	
105	Component missing	
106	Component incorrectly assembled	
107	Fuze well internal thread incorrect (Dwg C37-1-22)	
108	Internal or external rust	
109	Cover adapter thread incorrect	
110	Workmanship (3.8)	
Minor:	AQL 4.0 percent defective	
201	Protective finish missing or defective	

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4.2.3.3 Classification of defects (continued).

(g) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Components not clean
102	Dehydrator plug incorrect
103	VCI wrapping missing
104	Container closure incorrect
105	Strapping loose
106	Marking incorrect
107	Marking missing
108	Dehydrator plug color incorrect
Minor:	AQL 6.5 percent defective
201	Gross weight limitations in excess of container requirements
202	Components not packed in snug-fitting containers

4.2.4 Tests.

4.2.4.1 Shutter functioning and strength. The subassembly consisting of one cup, one shutter, and one shutter guide pin shall be supported and a load of at least 175 pounds shall be applied to the riveted end of the shutter guide pin by means of a hardened steel pin 0.100 - 0.005 inch in diameter with a pilot of 0.062 - 0.005 inch diameter by 0.03 inch long. The shutter guide pin shall withstand the minimum load. The subassembly plus one body and one shutter bearing pin shall be assembled. The resulting subassembly shall be supported and a load of at least 175 pounds shall be applied to the riveted end of the shutter bearing pin by means of the hardened steel pin described above. The shutter bearing pin shall withstand the minimum load. The arming plug knob shall be rotated to "SAFE" and "ARMED" positions to determine functioning of shutter.

4.2.4.2 Arming of the arming plug. The arming plug assembly shall be placed in a fixture and torque applied to the setting knob.

4.2.4.3 Mine handle assembly (following helium test). The mine handle assembly shall be tested using the Handle Pull Tester, Q86.

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4.2.4.4 Pressure plate assembly load. The pressure plate assembly load shall be tested using the Land Mine Pressure Plate Tester, Q87.

4.2.4.5 Internal air pressure of the arming plug. The arming plug assembly shall be placed in a fixture and subjected to an internal air pressure of 3 ± 1/2 pounds per square inch gage (psig). The assembly shall then be observed for leakage while submerged in water at a temperature of 75° - 5° F, to a depth of 1 ± 1/4 inch for a period of 20 ± 5 seconds.

4.2.4.6 Burster tube assembly. The burster tube assembly shall be tested using a Government approved helium leakage tester. If the burster tube is accepted, the letter "E" shall be stamped in ink on the hexagonal surface.

4.2.4.7 Agent compartment. With the mine assembled as shown on Drawing D37-1-34, the agent compartment shall be tested using a Government approved helium leakage tester. If the assembly is accepted, the letter "E" shall be stamped in ink near the filling port.

4.2.4.8 Void between the pressure place and fuze well (Dwg D37-1-58). The void between the pressure plate and fuze well shall be tested using a Government approved air leakage tester.

4.2.4.9 Burster cavity. The burster cavity shall be checked with a gage conforming to the maximum permissible dimension of the burster.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. The mine body and inert components shall be cleaned and dried in conformance with the applicable provisions of MIL-P-116. Cleaning method C-1 and the most applicable of drying methods listed therein shall be used.

5.2 Preservation and packaging.

5.2.1 Level B."

5.2,1.1 Mine body. Except for the inert components listed in 5.2.1.2, 5.2.1.3 and 5.2.1.4, all other inert components shall be assembled in place on the empty mine body as shown on Drawing D37-1-58. A blue dehydrator plug conforming to Drawing B37-1-65 shall be inserted with a torque of 40 ± 5 inch-pounds into the opening in the mine body that has been provided for the burster tube. Prior to emplacement of dehydrator plug, the counter bore and chamfered seal area shall be sparingly coated with type P-6 preservative, conforming to MIL-P-116. Caution shall be exercised so as not to contaminate the mine cavity or the adapter threads. The body shall not be packaged until the blue dehydrator plug has been inspected after at least 24 hours.

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5.2.1.2 Burster tube. Thirty-six burster tubes (Dwg C37-1-54) shall be packaged together in a snug-fitting W5c or W6c, style optional, fiberboard box conforming to PPP-B-636, with box closure with taping optional, to be as specified in the appendix thereto. The tubes shall be packaged in the above cited box in such manner as to prevent contact with each other, through the use of paperboard accordian-type separators or other internal cushioning devices. Prior to packaging, each burster tube shall be inclosed in a wrap of VCI-treated material conforming to type I, class 2, style A or B of MIL-P-3420.

5.2.1.3 Lead gaskets. Thirty gaskets (Dwg A37-1-16), that are to be used with the burster tubes, shall be wrapped together with grade B, Kraft wrapping paper conforming to UU-P-268 and packaged in either a snug-fitting paperboard folding, set-up, or paperboard metal-stayed box conforming to variety 1 of PPP-E-00566, variety 1 of PPP-B-676 or PPP-B-665 respectively, with closure to be in accordance with the appendix of the applicable container specification.

5.2.1.4 Steel internal retaining rings. Five hundred of the steel internal retaining rings (MS 9014-09) shall be packed in a snug-fitting paperboard folding, set-up, or paperboard metal-stayed box conforming to variety 1 of PPP-B-00565, variety 1 of PPP-3-676 or PPP-B-665, respectively, with closure to be in accordance with the appendix of the applicable container specification, The rings shall be packaged in the boxes in such manner as to prevent them from becoming entangled with each other. Prior to packaging retaining rings in box, an insert liner or bag fabricated from type I, class 2, style A or B, VCI-treated material conforming to MIL-P-3420 shall be placed therein. This liner or bag, when placed in the box, shall be of such size that after the rings are placed therein, the sides and ends of the liner shall protrude enough to permit a fold-over that will be sufficient to cover the rings.

5.3 Packing.

5.3.1 Level B.

5.3.1.1 Mine body. The empty mine body packaged as specified in 5.2.1 shall be packe as specified on Drawing D37-1-120. In addition, shipping containers shall be wrapped with waterproof wrapping, Kraft paper conforming to UU-P-271 and taped with one inch wide pressure sensitive, adhesive, waterproof tape conforming to PPP-T-76.

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5.3.1.2 Burster tube. Burster tubes packaged as specified in 5.2.1.2 shall be packed not to exceed gross weight limitations in either domestic-type nailed wood, wirebound wood, wood-cleated fiberboard, or wood-cleated plywood box conforming to PPP-B-621, PPP-B-585, PPP-B-591, or PPP-B-601, respectively; or in a V3c, style RSC fiberboard box conforming to PPP-B-636. Closure and strapping of the containers shall be in accordance with the applicable container specification and its appendix thereto, except that coated-finish strapping shall be used.

5.3.1.3 Lead gaskets. Lead gaskets packaged as specified in 5.2.1.3 shall be packed not to exceed gross weight limitations in a V3c, style RSC, fiberboard box conforming to PPP-B-636, with box closure and strapping to be in accordance with the appendix thereto except that coated-finish strapping shall be used.

5.3.1.4 Steel internal retaining rings. Steel internal retaining rings, packaged as specified in 5.2.1.4, shall be packed as specified in 5.3.1.3 except that in addition to closure requirements specified therein, all seams including the manufacturer's joint, shall be taped. Coated-finish strapping shall be used as box reinforcements.

5.3.1.5 Palletizaion. Palletization shall be in accordance with Drawing D37-1-117.

5.4 Markin In addition to any special marking required by the contract or order, shipping containers and palletized loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.2 Brazed components. Subcontractors should be advised of leakage factor for brazed components.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1345-A197

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contract, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 JAN 58 WHICH MAY BE USED.

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Edgewood Arsenal
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Edgewood Arsenal, Maryland 21010

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